MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7560-V

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic

See figures 1 and 2

Mounting position: Axis vertical, anode (water jacket) down. Tube shall be supported by anode flange only.

Weight: 110 pounds nominal

ABSOLUTE RATINGS: See note 1 (A)

Parameter:	F1	Ef	If	Ebb	Ec	eb	Pg	Pp	ik	T(seal)	tp	Du	Cooling
Unit:	MHz	V	---	kVdc	Vdc	kv	W	kW	a	°C	μs	---	---
Maximum:	30	15.7	---	50.0	-3,500	55.0	2,500	175	550	175	1,000	0.01	---
Minimum:	---	14.0	---	---	---	---	---	---	---	---	---	---	---
		Note 2	Note 3					Note 4	Note 2	Note 5			Note 4
TEST CONDITIONS:	---	14.5	---	---	---	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Not required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (see note 9)					
1266	Gas current	Ebb = 19.0 kVdc; Ec/Ib = 7.0 Adc; t = 300 sec (see note 10)	Ic	---	-1,200	μAdc
1261	Electrode voltage (1) (grid)	Ebb = 20.0 kVdc; Ec/Ib = 0.020 Adc	Ec (1)	-410	-585	Vdc
1261	Electrode voltage (1) (anode)	Ec = -150 Vdc; Ebb/Ib = 5.0 Adc	Ebb (1)	8.6	11.7	kVdc
1261	Electrode voltage (2) (anode)	Ec = -50 Vdc; Ebb/Ib = 5.0 Adc	Ebb (2)	4.6	6.9	kVdc
---	Positive grid drive	Ef = 15.0 V; Ebb = 6.0 kVdc; Ec = -400 Vdc; ec/ib = 300 a; Rp = 0; tp = 10 μs (max); prr = 50 (min) (see notes 2, 4, 11, and 12)	ec ic	--- ---	1,600 80	v a
---	Anode leakage	Ebb = 60 kVdc; Ec = -3,000 Vdc	Ib	---	15	mAdc
1261	Electrode voltage (2) (grid)	Ebb = 60 kVdc; Ec/ΔIb = 5 mAdc (see note 13)	Ec (2)	-1,250	-1,850	Vdc
1356	Grid pulse operation	Ef = 15.0 V; Ebb = 55 kVdc; Ec = -2,000 Vdc; ec/ib = 300 a (min); Rp = 140 ± 10%; tp = 2.5 μs (min); prr/Du = 0.0002 (min), 0.0005 (max); t = 15 minutes (see notes 2, 4, 11, 14, and 15)	---	---	---	---
	Quality conformance insp inspection, part 2 (see note 16)					
1301	Heater current		If	420	490	A
1331	Direct-interelectrode capacitance	No shield	Cgp Cgk Cpk	60 170 2.0	85 240 5.5	pF pF pF
Ⓐ ---	Power output	Ebb = 25 kVdc; Ec = -1,100 Vdc; rf drive/Ib = 20 Adc; Ic = 4 Adc ±10%; F = 13.6 MHz; t = 15 minutes (see notes 4, 5, 15, and 17)	Po	275	---	kW (useful)
	Quality conformance inspection, part 3					
---	Service-life guarantee	Within absolute ratings; t = 5,000 hours (see notes 1, 4, and 5)	---	---	---	---
---	Service-life end point	Grid pulse operation except Ef = 15.5 V (max) (see note 18)	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(A)	Periodic-check tests (see note 6)					
---	Vibration	No voltages (see note 7)	---	---	---	---
---	Shock	No voltages (see note 8)	---	---	---	---
---	Post-vibration and shock end points	Heater current Electrode voltage (1) (grid)	If Ec (1)	420 -410	490 -585	A Vdc

NOTES:

1. During fault conditions incited by tube or circuit instabilities, the tube shall be protected against the damaging effects of the very-high power available. The criterion is the speed in which the energy is removed. In addition, to the normal circuit breakers and overload relays, it is necessary that a high-speed electronic protective system, or equal, be used. Preferably this may be connected to short the rectifier output to ground in 20 microseconds or less during a fault. To limit surge currents it is further suggested that 5- to 10-ohms resistance be connected in series with each rectifier tube.

2. For ik greater than 350 a, Ef shall be 15.0 V minimum.

3. Heater surge current shall not exceed 1,200 amperes.

4. Anode is water cooled, resistivity of 100,000 ohms per cm^3 minimum. Waterflow shall be 40 gpm minimum. Pressure drop across jacket at 40 gpm is approximately 24 psi. For positive grid drive and grid pulse operation, 5- to 10-gpm waterflow is adequate, but for other tests 40 gpm shall be required. (See figure 4.) If other than water is used, the manufacturer's instructions on percentages, solution and flow rate shall be observed.

5. Forced-air cooling of bulb and seals is required. Approximately 500 cfm airflow should be directed primarily on heater seals and the main ceramic bulb. Airflow shall be distributed uniformly around the seals and ceramic insulators of the tube so as to maintain the seals and ceramic insulators at a uniform temperature not exceeding 175° C.

6. These tests and all other tests specified herein shall be performed when type 7560-V has not been previously furnished or has been previously furnished by the contractor to the Government but there have been subsequent changes in basic tube design or construction, or production has been discontinued for a 3-year period. A minimum of three tubes shall be tested. A failure of any of the tests shall be cause for rejection of the sample. Three complete copies of the test reports shall be submitted to the procuring activity with approval prior to production of the tubes to be furnished under the contract.

7. The tube shall withstand, without permanent damage, vibration at 25 Hz plus or minus 2 Hz with a constant total displacement of approximately 0.020 inches with the tube axis perpendicular to the direction of motion of the vibration table. The duration of the test shall be 2 minutes minimum.

8. The tube shall be firmly mounted to a platform and dropped on a sand bed to give a pulse duration of about 11 milliseconds with a peak value of the shock pulse greater than 5G. The tube shall withstand, without permanent damage, 10 impact shocks as follows:

   Vertically upright - 5 shocks.
   Vertically inverted - 5 shocks.

9. These tests shall be performed on each tube. A failure of any test shall be cause for rejection of the tube.

10. After the specified time, note grid current as (1)Ic. Then increase -Ec to obtain 0.020 Adc anode current at Ebb = 20.0 kVdc (per following electrode voltage (1) (grid test)) and note grid current as (2)Ic. Record gas current as Ic = (1)Ic-(2)Ic.

11. Symbol ec is peak positive grid voltage. Pulse current and voltage values shall be determined *using peak-reading voltmeter, oscilloscope, or other suitable indicating means.*

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NOTES: - Continued

12. No resistance shall be included in the anode circuit other than a current viewing resistor.

13. Starting with Ec = -3,000 Vdc from anode leakage test, Ec is made less negative until Ib increases by the specified amount.

14. This test may be preceded by high-voltage aging at gradually increased anode voltage (a) with heater cold, then (b) with Ef = 15.0 V, and by grid pulse aging as required. Total time for pre-aging run shall be 1-hour maximum.

15. The tube shall operate for the specified time without kicking out the overload relay, within six attempts.

16. Test 1 tube selected at random from each group of 10 tubes to be delivered. If one tube fails the specified end points, the failed test shall become a part of quality conformance inspection, part 1, on all groups of 10 tubes in process. After 3 groups of 10 tubes have successfully passed the failed test, the test shall revert to quality conformance inspection, part 2.

17. The tube shall be tested in power amplifier.

18. The tube shall be tested in the same manner as for grid pulse operation test, except that filament voltage may be adjusted to the specified maximum value, if necessary, to obtain the required cathode emission.

Custodian:
Air Force - 17

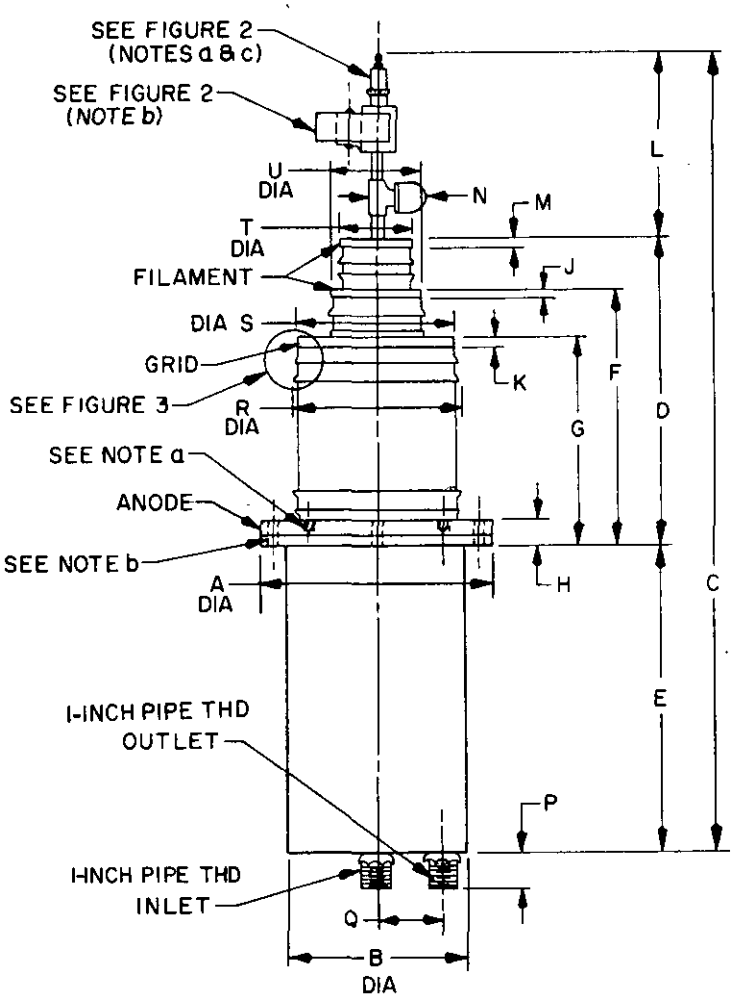
Review activities:
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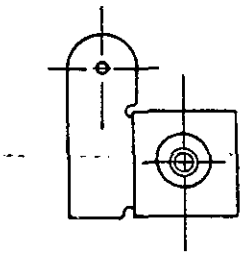


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3 (periodic check)		
A	9.94 (252.48)	10.06 (255.52)
B		8.13 (206.50)
C		36.50 (927.10)
D	13.08 (332.23)	13.68 (347.47)
E	12.56 (319.02)	13.06 (331.72)
F	10.97 (278.64)	11.41 (289.81)
G	9.00 (228.60)	9.38 (238.25)
H	1.07 (27.18)	1.19 (30.23)
J	.313 (7.95)	
K	.500 (12.70)	
L		8.50 (215.90)
M	.313 (7.95)	
N		2.50 (63.50)
P		1.50 (38.10)
Q	2.81 (71.37)	3.19 (81.03)
R		7.38 (187.45)
S	6.907 (175.44)	6.969 (177.01)
T	3.282 (83.36)	3.344 (84.94)
U	3.969 (100.81)	4.031 (102.39)

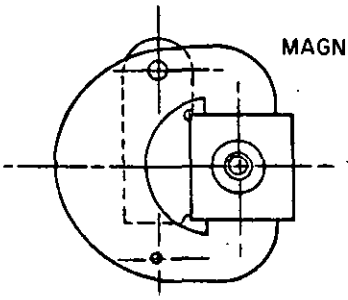
NOTES:
a. Four holes ¼-20 UNC-2B equally spaced on 8.19 (208.03) ± .03 (.76) B.C. for lifting.
b. Four through holes .290 (7.37) dia on 9.16 (232.66) ± .03 (.76) B.C. for mounting, located approximately 45° from tapped holes.

FIGURE 1. Outline drawing of electron tube type 7560-V.

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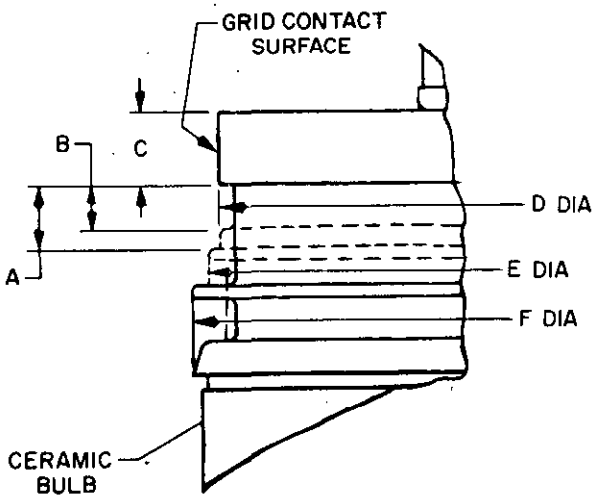
TOP VIEW- ION SORPTION PUMP



VIEW SHOWING MAGNET LOCATION

NOTES:
a. Ion sorption pump may be oriented approximately 90° or 180° from position shown with reference to outlet water connection.
b. Magnet is required for operation of ion sorption pump. Install magnet as shown in view above. Remove magnet for shipping and handling of the tube. The magnet is supplied as a part of each 7560-V tube.
c. Alternate ion sorption pump and magnet configurations may be used, provided that all of the following conditions are met:
   (1) Rated pump speed: 1 liter per second minimum.
   (2) Cathode operating potential: 3kVdc ±10%.
   (3) The 8.50 (215.90 mm) maximum dimension of figure 1 is not exceeded.
   (4) Dimensions are such as to permit operation with the power supply connector as shown on figure 5.

FIGURE 2. Ion sorption pump and magnet location.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3 (periodic check)		
A	.344 (8.74)	
B	.187 (4.75)	
C	.500 (12.70)	
D		7.04 (178.82)
E		7.19 (182.63)
F		7.38 (187.45)

FIGURE 3. Detail of grid terminal.

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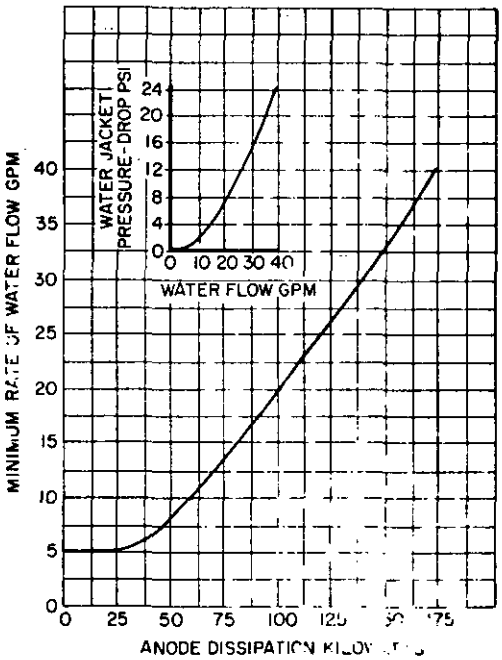
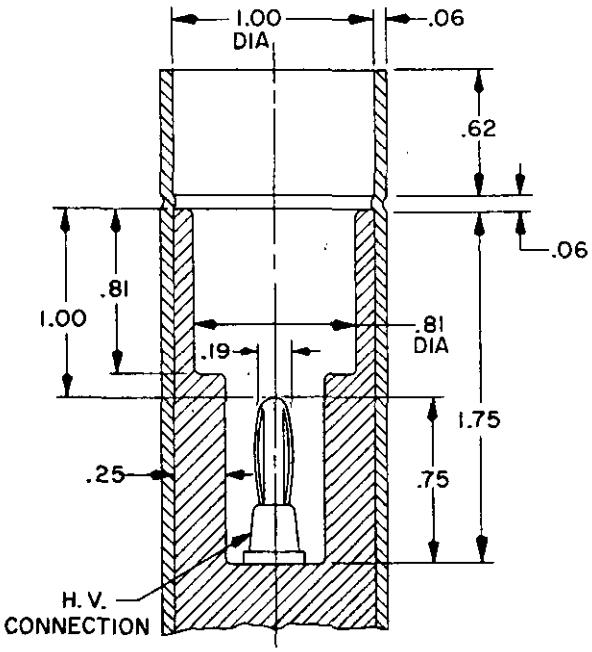


FIGURE 4. Water cooling characteristics.



INCHES	MM
.06	1.52
.19	4.83
.25	6.35
.62	15.75
.75	19.05
.81	20.57
1.00	25.40
1.75	44.45

NOTE:
a. Unless otherwise specified, tolerance is ±.03 (.76 mm).

FIGURE 5. Power supply connector for ion sorption pump.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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